

# A proof of the conjectured recurrence for OEIS A306948

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## Abstract

OEIS A306948 carries a conjectured linear recurrence with polynomial coefficients, of order 4. The entry gives no generating function, but it does give the sequence as a hypergeometric sum. Creative telescoping produces, from the summand alone, a certificate identity that is checked exactly rather than trusted. What it does not produce here is a homogeneous recurrence: the summand does not vanish at the ends of the range, and the range itself moves with  $n$ , so summing the certificate leaves boundary and range-correction terms. We keep them. They are a finite, explicit expression  $h(n)$ , and the result is an inhomogeneous recurrence  $\sum_i \sigma_i(n)a(n+i) = h(n)$ . Since  $h$  is a hypergeometric term, the first-order operator  $M = N - h(n+1)/h(n)$  annihilates it, and  $ML$  annihilates  $a$ . The conjectured operator is then shown to be a left multiple of the resulting annihilator by division in the Ore algebra  $\mathbb{Q}(n)[N]$ .

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## 1 The sequence and the conjecture

OEIS A306948 is “Expansion of e.g.f.  $(1+x)^{\log(1+x)\exp(x)}$ ”. It has offset 0 and begins

$$a(0), \dots, a(7) = 0, 1, 3, 5, 8, 9, 19, -15, \dots$$

The entry states, not as a conjecture,

$$a(n) = \text{Sum}_{k=1..n} (-1)^{k-1} \binom{n}{k} (n-k+1) (k-1)!.$$

so that, writing  $F(n, k)$  for the summand,

$$a(n) = \sum_{k=1}^n F(n, k), \quad F(n, k) = (-1)^{k-1} (-k+n+1) \binom{n}{k} (k-1)!. \quad (1)$$

This is the only input the proof takes from the entry, and it was checked against the entry’s own DATA before use.

The entry separately carries the following comment:

**Conjecture:** D-finite with recurrence  $a(n) + (n-5)a(n-1) + (-3n+10)a(n-2) + 3(n-3)a(n-3) + (-n+3)a(n-4) = 0$ . - R. J. Mathar, Aug 20 2021

As of the “Last modified” line on the live entry (2021-10-29, revision 13) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

## 2 Telescoping over a range that does not cooperate

Let  $N$  denote the shift in  $n$ ,  $(Nf)(n) = f(n+1)$ , and  $\Delta_k$  the forward difference in  $k$ . Creative telescoping looks for rational functions  $\sigma_0, \dots, \sigma_r$  of  $n$ , not all zero, and a rational  $R(n, k)$  with

$$\sum_{i=0}^r \sigma_i(n) F(n+i, k) = \Delta_k [R(n, k) F(n, k)]. \quad (2)$$

Divided through by  $F(n, k)$  this is an identity between rational functions, because the quotients  $F(n+i, k)/F(n, k)$  and  $F(n, k+1)/F(n, k)$  are rational; so a proposed pair  $(\sigma, R)$  can be *checked*, exactly, whatever produced it. For A306948 such a pair exists with  $r = 2$ :

$$\sigma_0(n) = 1, \quad \sigma_1(n) = \frac{1-n}{n+1}, \quad \sigma_2(n) = -\frac{1}{n+1}, \quad R(n, k) = \frac{k^2 - kn - 3k}{k^3 - 3k^2n - 4k^2 + 3kn^2 + 8kn + 5k - n^3}$$

The usual next step sums (2) over all  $k$  and concludes  $\sum_i \sigma_i(n) a(n+i) = 0$ . That step needs  $F(n, k)$  to vanish for every  $k$  outside the range in (1), and here it does not. Summing over a finite window therefore leaves something behind, and the point of this note is that what it leaves behind is computable.

**Proposition 1.** *Let  $L_0$  and  $H_0$  be the lower and upper summation limits in (1), each non-decreasing in  $n$ , and put  $L = L_0(n)$  and  $H = H_0(n+r)$ , so that the window  $L \leq k \leq H$  contains the range of every  $a(n+i)$ ,  $0 \leq i \leq r$ . Write  $G(n, k) = R(n, k)F(n, k)$  and suppose  $G$  is finite at every integer  $k$  in  $[L, H+1]$ . Then*

$$\sum_{i=0}^r \sigma_i(n) a(n+i) = h(n), \quad h(n) = G(n, H+1) - G(n, L) - \sum_{i=0}^r \sigma_i(n) \sum_{\substack{L \leq k \leq H \\ k \notin [L_0(n+i), H_0(n+i)]}} F(n+i, k).$$

*Proof.* Sum (2) over  $k = L, \dots, H$ . The right-hand side telescopes to  $G(n, H+1) - G(n, L)$ , every intermediate value being finite by hypothesis. On the left,  $\sum_{k=L}^H F(n+i, k)$  differs from  $a(n+i) = \sum_{k=L_0(n+i)}^{H_0(n+i)} F(n+i, k)$  exactly by the terms of the window that lie outside that sequence's own range, which is the inner sum above. Rearranging gives the statement; both correction ranges are finite because  $L_0$  and  $H_0$  are non-decreasing and  $i \leq r$ .  $\square$

For A306948 this gives

$$h(n) = \frac{-n-2}{n+1}. \quad (3)$$

## 3 Clearing the inhomogeneity, and dividing

The inhomogeneity (3) is a hypergeometric term: its shift quotient is

$$\frac{h(n+1)}{h(n)} = \frac{n^2 + 4n + 3}{n^2 + 4n + 4},$$

a rational function of  $n$ . Hence  $M = N - h(n+1)/h(n)$  satisfies  $M(h) = 0$ , and writing  $L_0 = \sum_i \sigma_i(n) N^i$  for the telescoped operator, Proposition 1 reads  $L_0(a) = h$ , so

$$L = ML_0 \quad \text{satisfies} \quad L(a) = M(L_0(a)) = M(h) = 0.$$

One extra order buys the homogeneity. Explicitly,  $L = \left( \frac{-n^2-4n-3}{n^2+4n+4} \right) + \left( \frac{2n^2+6n+1}{n^2+4n+4} \right) N + \left( \frac{-n^2-n+3}{n^2+4n+4} \right) N^2 + \left( -\frac{1}{n+2} \right) N^3$ .

**Lemma 2.** *If  $C = QL$  in  $\mathcal{A} = \mathbb{Q}(n)[N]$ , where  $Nq(n) = q(n+1)N$ , and  $L(a) = 0$ , then  $C(a) = 0$  at every  $n$  where the coefficients of  $Q$  are defined.*

*Proof.*  $C(a) = (QL)(a) = Q(L(a)) = Q(0) = 0$ , and the coefficients of  $Q$  are rational functions of  $n$ , so the step is valid away from their poles.  $\square$

Writing the conjectured recurrence as an operator  $C$  and dividing by  $L$  on the right leaves remainder 0, with quotient

$$Q = \left( \frac{n^2 + 4n + 4}{n + 3} \right) + (-n - 3) N.$$

The coefficients of  $Q$  are rational functions of  $n$ ; their only integer poles are  $n \in \{-3\}$ , below the range claimed here.

**Theorem 3.** *The conjectured recurrence*

$$1a(n) + (n-5)a(n-1) + (10-3n)a(n-2) + 3(n-3)a(n-3) + (3-n)a(n-4) = 0$$

*holds for every  $n \geq 4$ .*

*Proof.* Proposition 1 gives the inhomogeneous recurrence, the operator  $L$  of the previous section annihilates  $a$ , and Lemma 2 applied to  $C = QL$  carries that to the conjectured operator. The integer poles of  $Q$  all lie below  $n = 4$ .  $\square$

## 4 Verification

Every number above is an output of a script written separately from this note, and four independent checks were run.

First, the certificate was verified rather than trusted: (2) was divided by  $F(n, k)$  and the difference of the two sides cancelled to 0 as a rational function of  $n$  and  $k$ .

Second — and this is the check that matters most here, because the boundary bookkeeping is where such an argument goes wrong — the inhomogeneous recurrence of Proposition 1 was evaluated on the entry's own terms: for each  $n$  in range the left side was formed from the published values and compared against (3), and the two agreed exactly. An error in the correction terms would not survive this.

Third, the factorisation was confirmed by multiplying back:  $QL$  was expanded in  $\mathcal{A}$  and its coefficients cancelled against those of  $C$  to zero.

Fourth, the conjecture itself was evaluated on the published terms, in exact integer arithmetic and without reference to any operator: for each  $n$  in range the sum  $\sum_i p_i(n)a(n-i)$  was formed from the entry's own values and found to vanish, for all 21 values of  $n$  from  $n = 4$  upward. The formula (1) was likewise checked term by term against the DATA section before being used.

## References

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